

Climate Risk Premium Model

Samcheok Coal Asset

Quantifying transition and physical climate risk
for project finance

PLANiT Institute

February 2026

1

Baseline NPV: \$4,629M

Under current policy, the Samcheok asset is rated AA with strong debt service coverage (DSCR 2.56).

2

Enhanced phase-down: -\$1,891M (CC rated)

Korea's 11th Basic Plan for Electricity triggers a \$6.5 billion value swing — the single largest risk driver.

3

Climate risk premium: up to 1,020 bps

Under aggressive transition, lenders would demand 10.2% additional yield over risk-free benchmarks.

4

Transition risk dominates physical risk by 103:1

Policy-driven cash flow erosion overwhelms direct physical damage across every scenario tested.

5

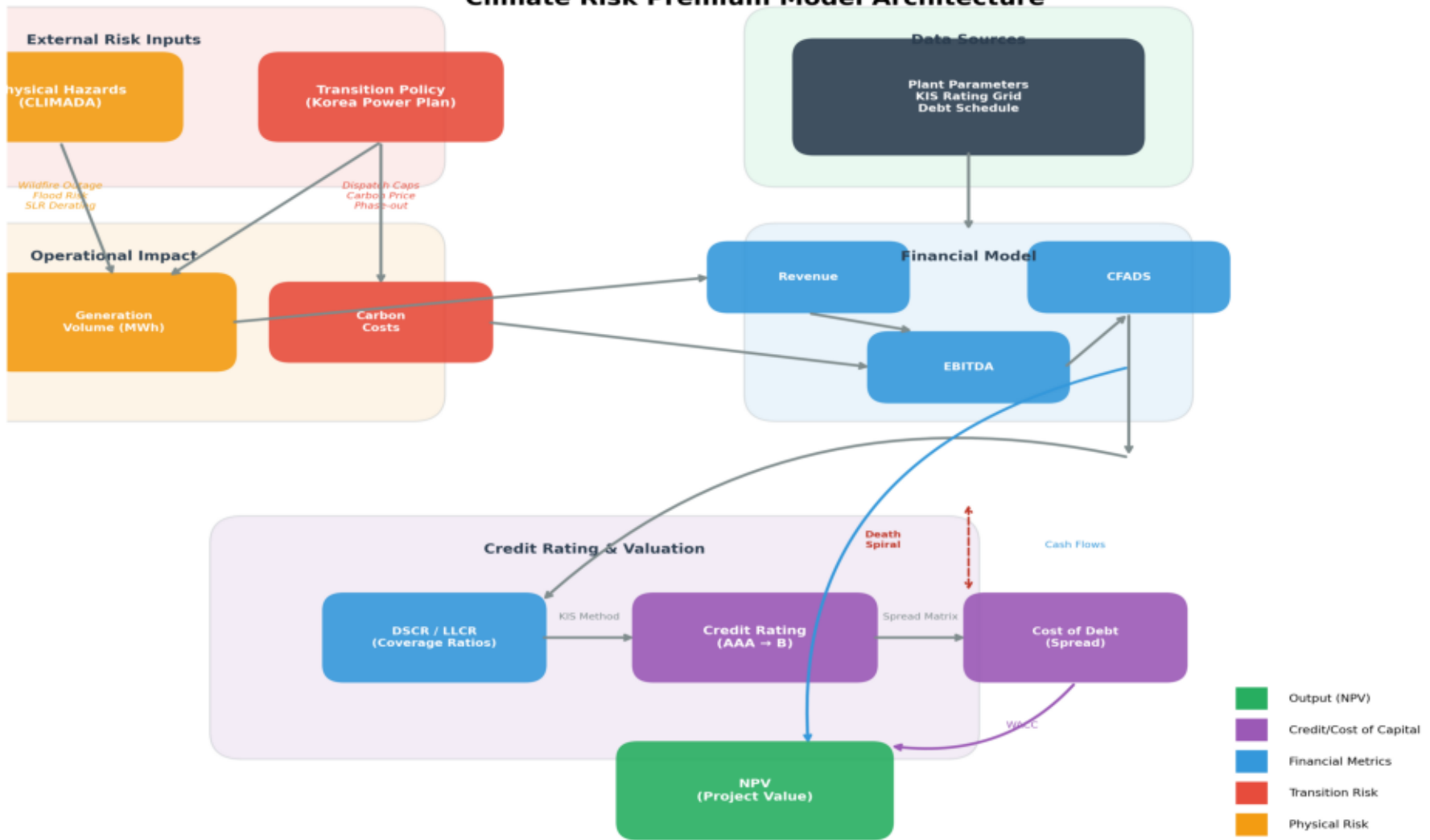
Results robust across 100% of stress grid

All 648 deterministic grid points confirm transition dominance; no reversal found.

Samcheok Blue Power Plant is a 2.1 GW supercritical coal-fired generation facility located in Gangwon Province, South Korea. Commissioned in 2017 with a 30-year design life, the plant is one of the youngest and most efficient coal assets in Korea's fleet. It represents a critical test case for stranded-asset risk under the national energy transition framework.

Parameter	Value	Parameter	Value
Capacity	2,100 MW	CAPEX	\$3,550 M
Technology	Supercritical coal	Debt / Equity	70% / 30%
Heat rate	8.8 MMBtu/MWh	Interest rate	5.0%
Capacity factor	85%	Tax rate	25%
Emissions	0.82 tCO ₂ /MWh	Design life	30 years
Efficiency	39%	Base electricity price	\$65/MWh

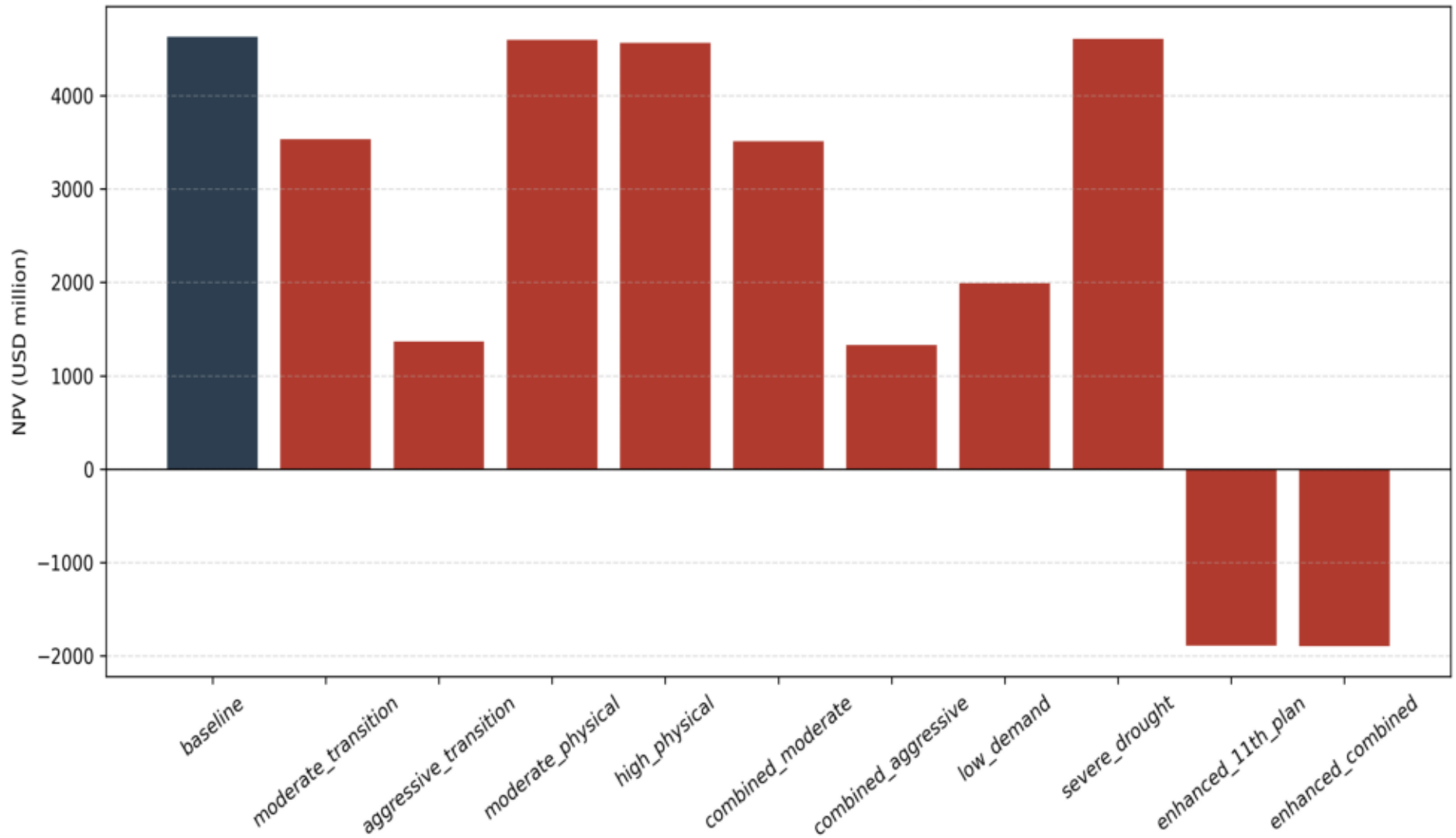
Climate Risk Premium Model Architecture



Scenario Inputs → Operational Impact → Financial Model → Credit Rating → Climate Risk Premium

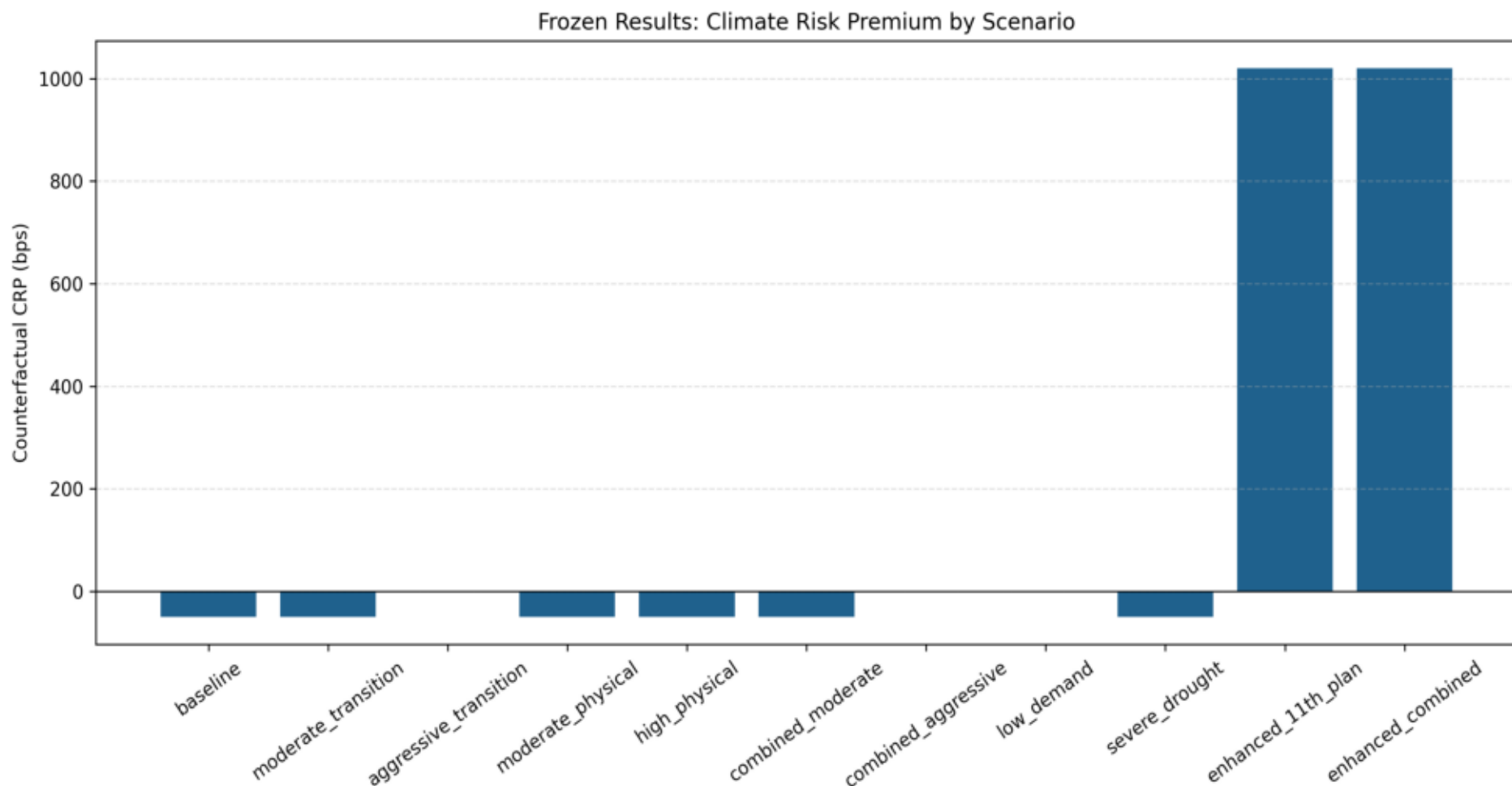
Scenario	Type	Key Driver	NPV (\$M)	Rating
Baseline	Reference	Current policy	4,629	AA
Moderate transition	Transition	Dispatch penalty 15%	3,534	AA
Aggressive transition	Transition	Dispatch penalty 40%	1,369	A
Moderate physical	Physical	1.5× climate mult.	4,599	AA
High physical	Physical	2.5× climate mult.	4,566	AA
Combined moderate	Combined	Trans. + phys. moderate	3,513	AA
Combined aggressive	Combined	Trans. + phys. aggressive	1,330	A
Low demand	Stress	Demand shock –25%	1,994	A
Severe drought	Stress	Water stress penalty	4,609	AA
Enhanced 11th Plan	Enhanced	Full phase-down schedule	-1,891	CC
Enhanced combined	Enhanced	Phase-down + phys. high	-1,896	CC

Frozen Results: Scenario NPV Comparison



Baseline: \$4,629M (AA)

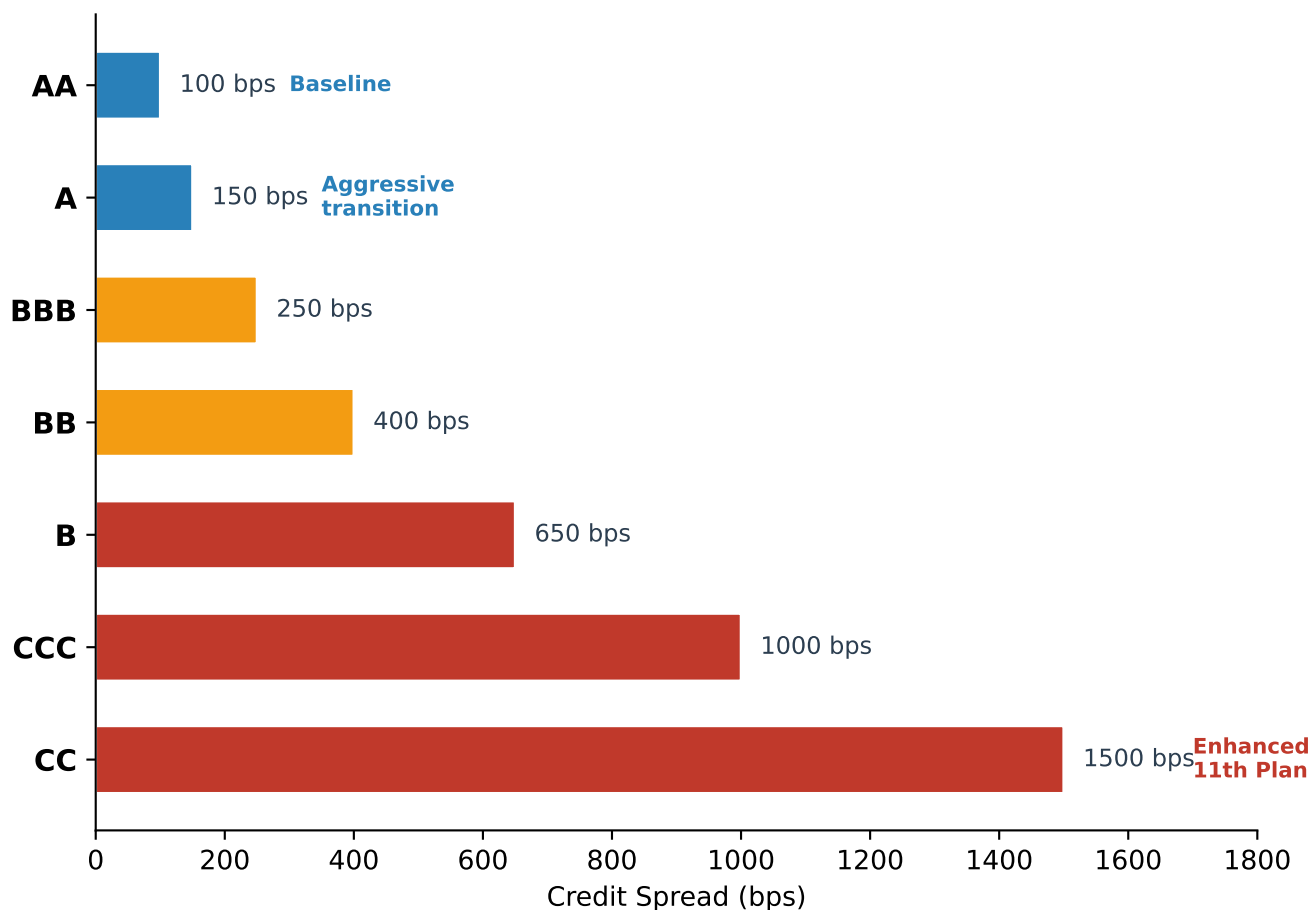
Enhanced 11th Plan: -\$1,891M (CC)



CRP = the additional cost of capital demanded by lenders to compensate for climate-related credit deterioration, measured as the spread differential between the climate-adjusted and baseline credit ratings.

Peak CRP: 1,020 bps = 10.2% additional financing cost

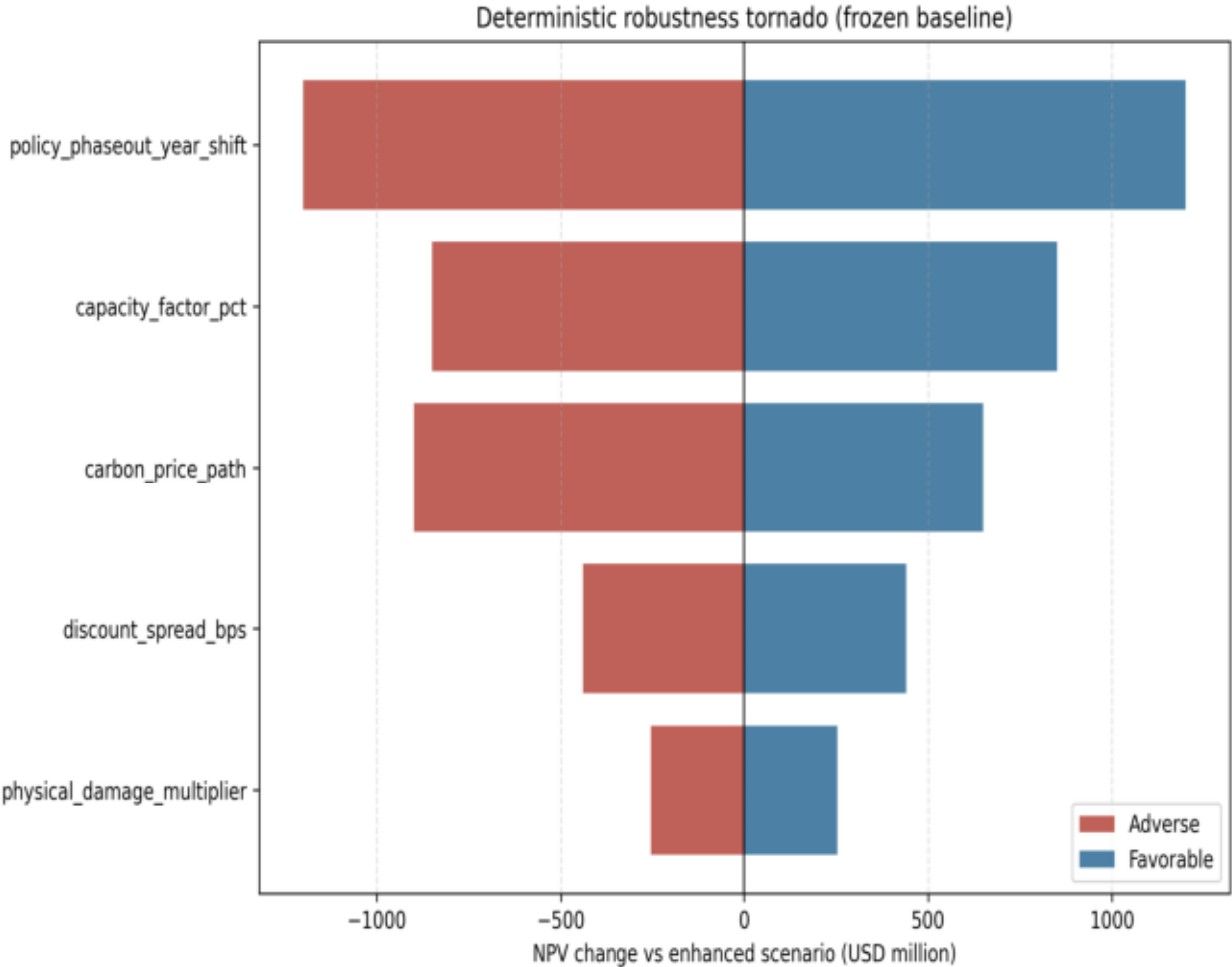
Rating Scale & Indicative Spreads



The 'death spiral' mechanism:

1. Carbon pricing & dispatch penalties erode operating margins
2. Negative EBITDA triggers credit downgrade (AA → CC)
3. Higher spreads increase financing costs, further eroding cash flow
4. Refinancing becomes prohibitively expensive or impossible

AA	A	BBB	BB	B	CCC	CC
100	150	250	400	650	1,000	1,500



Stress Range
-3,091M to 691M

Best Case
-\$691M (still negative)

Worst Case
-\$3,091M

Dominance Ratio
103:1 (trans. vs phys.)

Grid Coverage
100% transition dominance

Physical damage multiplier
is the smallest sensitivity
bar in every test.

Key Takeaways

1

Transition-policy clarity is financially consequential at the project level.

A single regulatory decision (the 11th Basic Plan) drives a \$6.5B value swing. Investors need scenario-conditioned risk metrics, not point estimates.

2

Disclosure governance matters as much as model sophistication.

Credit rating transitions from AA to CC are driven by policy assumptions, not model complexity. Transparent scenario design is essential.

3

Transition planning and financing design should be discussed jointly.

The 'death spiral' from carbon costs to distressed ratings to prohibitive spreads means energy transition policy and financial structuring are inseparable.